Name: _____

TOPIC TEST 2

Chapter 6: Time series analysis

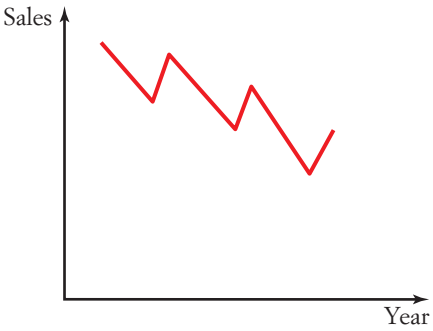
- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 8 multiple-choice questions (8 marks)
- Part B: 6 short-answer questions (34 marks)
- Part C: 2 analysis questions (20 marks)

Part A

Multiple-choice questions

8 multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

1 The time series graph shown below displays:



A Random fluctuations only

B Seasonal variation

C Cyclic variation

D A secular positive trend

E A secular negative trend

[1 mark]

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2 The time series graph shown below displays:



A Random fluctuations only

B Seasonal variation

C Cyclic variation

D A secular positive trend

E A secular negative trend

[1 mark]

3 The seasonal indices (SI) for sales of wetsuits by a surf factory outlet are shown in the table below.

Quarter	1	2	3	4
SI	1.23	0.97	0.66	1.14

The actual numbers of wetsuits sold in the second and third quarters, respectively, were 100 and 261. The seasonally adjusted sales figures are closest to:

A 100, 261

B 103, 395

C 395, 103

D 395, 269

E 269, 395

[1 mark]

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4 The seasonal indices (SI) of the sales from a retail outlet are shown in the table below.

Quarter	SI
1	0.1
2	1.4
3	1.3
4	1.2

For the first quarter of the year, the deseasonalised sales were a low $100 000. The actual sales figure for the first quarter was:

A $10 000

B $100 000

C $7 142 857

D $7 692 308

E $1 000 000

[1 mark]

5 For the points (2, 1), (4, 2.5), (7, 5.5), (10, 7), the least squares regression line can be expressed as:

A $y = -1.27t + 0.68$

B $y = 0.68t - 1.27$

C $y = -0.78t + 0.46$

D $y = 0.78t - 0.46$

E $y = 0.75t - 0.5$

[1 mark]

6 For the following time series situation, which is most likely to apply?

The oxygen levels of a patient with chronic lung failure.

A Positive secular trend

B Negative secular trend

C Seasonal pattern

D Cyclic pattern

E Random pattern

[1 mark]

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7 A manufacturer of recipe books has determined the quarterly proportions shown below describing the shop's quarterly sales projected figures for 3 years.

Year	Q1	Q2	Q3	Q4
2014	0.70	0.86	1.08	1.36
2015	0.64	0.85	1.23	1.28
2016	0.53	0.91	1.45	1.11

The seasonal index for the third quarter is:

A 1.35

B 0.71

C 0.87

D 1.25

E 0.62

[1 mark]

8 A teacher wanted to analyse the number of student absences in his classes. He listed the daily average absentee rates for Year 9 students for each term over a 2-year period and calculated these seasonal indices:

Year	Q1	Q2	Q3	Q4
Seasonalised index	0.65	0.90	1.06	1.39

If the equation describing the deseasonalised data is:

$$Absences = 1.7 \times Term + 26$$

the predicted actual number of absences in the first term of the third year is:

A 41

B 27

C 64

D 28

E 18

[1 mark]

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Part B

Short-answer questions

6 short-answer questions
34 marks
Show your working where appropriate.

1 **a** Use the three-point moving averages method to smooth the following data set. Plot the resulting graph.

t	1	2	3	4	5	6	7	8	9	10
y	3	5	6	10	9	16	12	16	17	20

b What type of trend is this?

[5 marks]

2 The following table shows the interest rates of a personal loan during a volatile period in the economy.

Day t	1	2	3	4	5	6	7	8	9	10
Rate y (%)	3.23	5.55	6.36	10.10	9.10	16.20	12.10	16.00	17.25	10.10

a Plot a time series plot and fit a three-median smoothing line to it.

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b What type of trend is this?

[5 marks]

3 Draw a time series plot and describe its features for the following data set.

t	1	2	3	4	5	6	7	8
y	1	1	6	2	11	4	17	16

[4 marks]

4 An asthma patient records her peak flow lung capacity, y, during an exacerbation of her disease. She records this data at the same time each day, t, and constructs the following table.

t	Mon	Tue	Wed	Thu	Fri	Sat	Sun
y	210	220	200	280	300	320	355

a Draw a time series plot and describe its features.

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b Find the least squares regression line and add it to your time series plot.

[6 marks]

5 Staff in a small country hospital tabulate the number of deaths recorded over a period of 3 years as shown in the table below.

Year	Q1	Q2	Q3	Q4
2011	3	4	2	5
2012	1	5	2	4
2013	2	2	5	6

a Create a time series plot using the data given.

b Using technology, find the least squares regression line and add it to your time series plot.

c State the equation of the least squares regression line.

d Use the equation to predict the number of deaths in the small country hospital in the first quarter of 2014.

[8 marks]

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6 A street vendor in New York City is trying to predict his sales of waffles for 2014. The table below shows his sales figures for the years 2008 to 2013.

Year *t*	2008	2009	2010	2011	2012	2013
Sales *y* ($1000s)	10	11	12	10	12	15

Using a two-point moving median with *centring*, smooth the data and use it to predict his sales for 2014.

[6 marks]

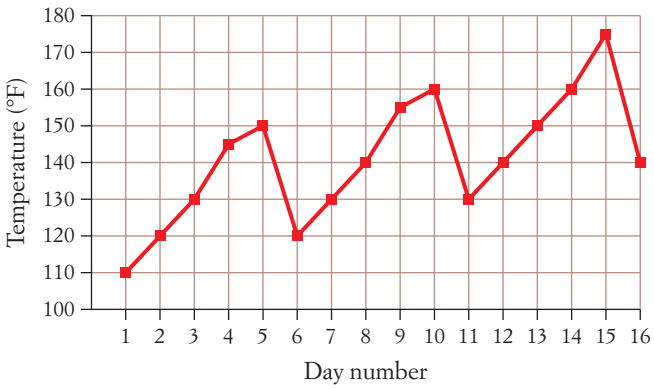
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Part C

Analysis questions

2 analysis questions
20 marks
Show your working where appropriate.

1 The temperature in a greenhouse is measured daily and the results are displayed in the graph below.



a Smooth the time series data using a three-point moving average.

b Use the smoothed data to predict the temperature for day 17.

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c Smooth the time series data using a four-point moving average.

d Describe the trend shown.

[10 marks]

2 The number of aces recorded during a particular timed four-set game at Wimbledon was recorded by an Australian commentator for each of the years from 2010 to 2012.

Year	Aces recorded set 1	Aces recorded set 2	Aces recorded set 3	Aces recorded set 4
2010	10	70	90	10
2011	10	80	90	40
2012	20	90	100	10

a Calculate the seasonal index for each set.

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b Deseasonalise the data.

c Plot the original and the deseasonalised time series data.

[10 marks]

[Total marks: 62]

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Answers

Part A – Multiple-choice questions **[1 mark each]**

1 The time series data over the year shown is moving in an overall negative direction.

$\therefore$ **E**

2 The time series data shown is for a period of 3 months. If we were to divide the graph into three sections, one for each month, we can see that there is a distinct pattern that repeats each month.

$\therefore$ **B**

3 $\text{Deseasonalised value} = \dfrac{\text{Actual value}}{\text{Seasonal index}}$

$$\text{D.V. (2nd quarter)} = \frac{100}{0.97} = 103.09\ldots \approx 103$$

$$\text{D.V. (3rd quarter)} = \frac{261}{0.66} = 395.\dot{4}\dot{5}\ldots \approx 395$$

$\therefore$ **B**

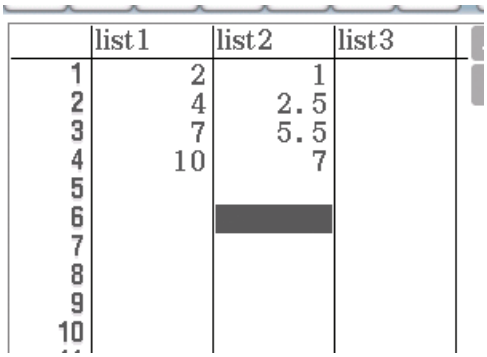
4 $\text{Actual value} = \text{Seasonal index} \times \text{Deseasonalised value}$

$= \$100\,000 \times 0.1$

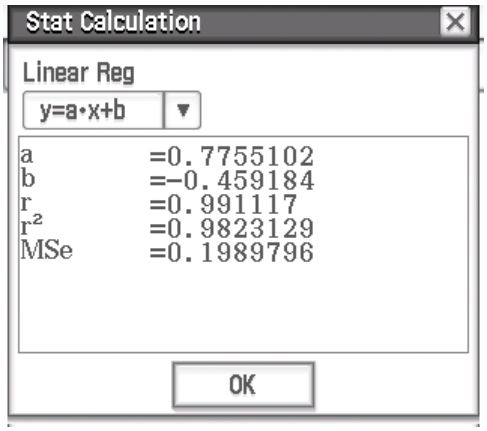
$= \$10\,000$

$\therefore$ **A**

5 CAS required to plot the points and create the least squares regression line, as shown below.



	list1	list2	list3
1	2	1	
2	4	2.5	
3	7	5.5	
4	10	7	
5			
6			
7			
8			
9			
10			



$\therefore$ **D**

6 The lungs of a person with chronic lung failure will gradually deplete and oxygen levels will drop as lung function worsens. Therefore, a graph of oxygen levels will be pointing downwards; i.e. a negative secular trend.

$\therefore$ **B**

7 Seasonal index for third quarter

$$= \frac{1.08 + 1.23 + 1.45}{3}$$

$= 1.25\dot{3}$

≈ 1.25

$\therefore$ **D**

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8 $Absences = 1.7 \times Term + 26$

$= 1.7 \times 1 + 26$

$= 27.7$

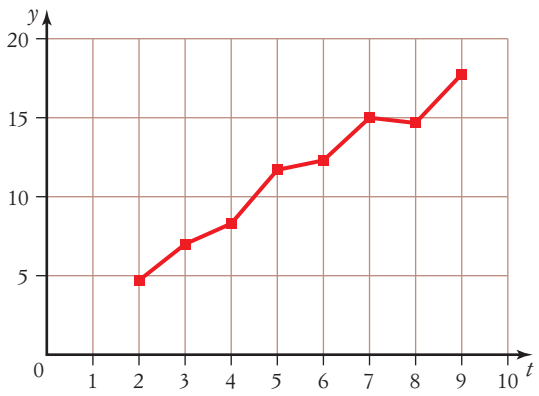
≈ 28

$\therefore$ **D**

Part B – Short-answer questions

1 a

t	1	2	3	4	5	6	7	8	9	10
y	3	5	6	10	9	16	12	16	17	20
		$\left(\frac{3+5+6}{3}\right)$	$\left(\frac{5+6+10}{3}\right)$	$\left(\frac{6+10+9}{3}\right)$	$\left(\frac{10+9+16}{3}\right)$	$\left(\frac{9+16+12}{3}\right)$	$\left(\frac{12+16+17}{3}\right)$	$\left(\frac{16+12+16}{3}\right)$	$\left(\frac{16+17+20}{3}\right)$	
		$= 4.7$	$= 7.0$	$= 8.3$	$= 11.7$	$= 12.3$	$= 15.0$	$= 14.7$	$= 17.7$	

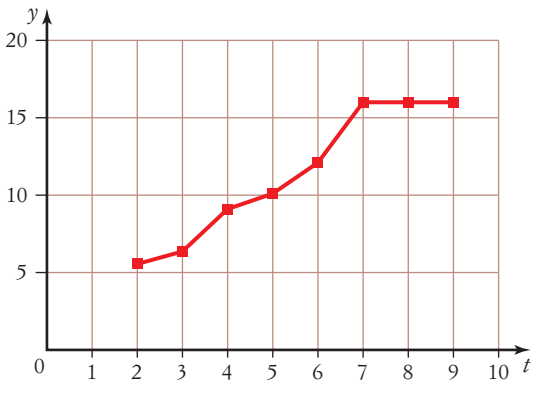


[4 marks]

b Positive secular trend **[1 mark]**

2 a

t	1	2	3	4	5	6	7	8	9	10
y ($)	3.23	5.55	6.36	10.10	9.10	16.20	12.10	16.00	17.25	10.10
		Med (3.23, 5.55, 6.36)	Med (5.55, 6.36, 10.10)	Med (6.36, 10.10, 9.10)	Med (10.10, 9.10, 16.20)	Med (9.10, 16.20, 12.10)	Med (16.20, 12.10, 16.00)	Med (12.10, 16.00, 17.25)	Med (16.00, 17.25, 10.10)	
		$= 5.55$	$= 6.36$	$= 9.10$	$= 10.10$	$= 12.10$	$= 16.00$	$= 16.00$	$= 16.00$	

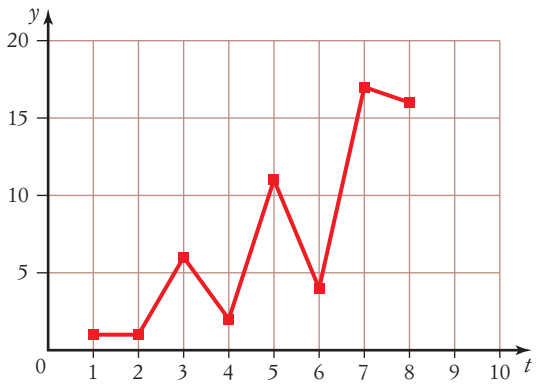


[4 marks]

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b Positive secular trend **[1 mark]**

3



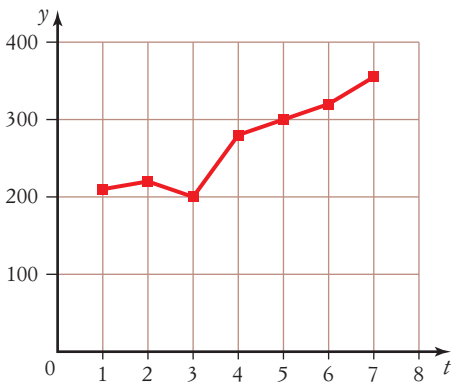
[3 marks]

Cyclic trend peaking approximately every second point **[1 mark]**

4

t	1	2	3	4	5	6	7
y	210	220	200	280	300	320	355

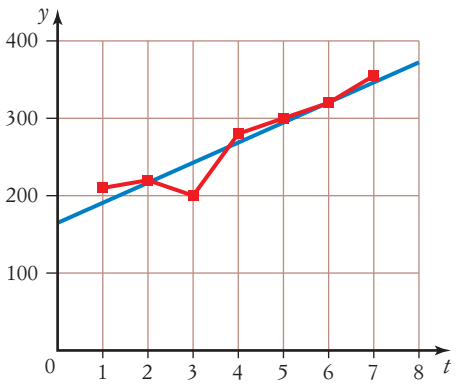
a



Plot shows patient's peak flow is improving over the 7 days. **[3 marks]**

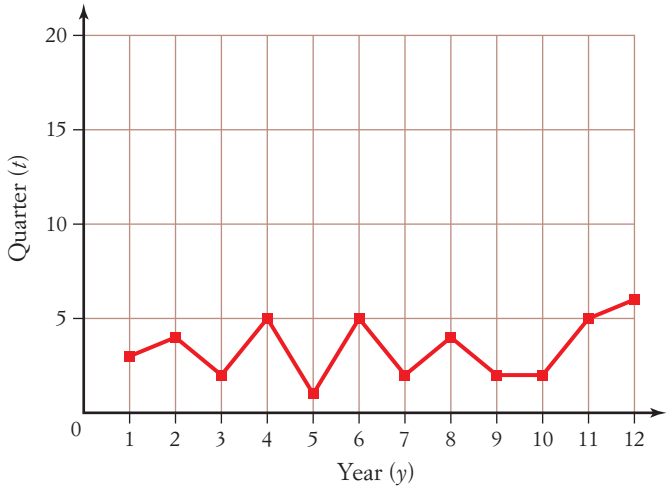
b Equation of least squares regression line is $y = 26.25t + 164.29$. **[1 mark]**

Addition to the time series plot of the least squares regression line is shown below **[2 marks]**



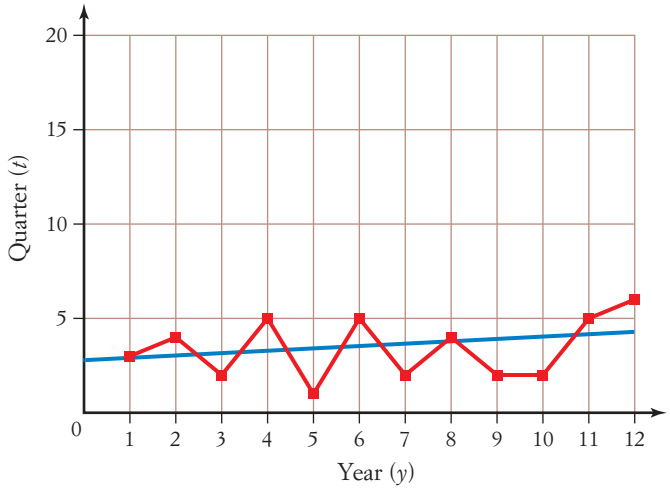
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5 a



[3 marks]

b



[2 marks]

c $y = 0.12t + 2.67$ **[1 mark]**

d At $t = 13$, $y = 0.12 \times 13 + 2.67 = 4.23$

Round down to 4 deaths.

Predicted number of deaths for the first quarter of 2014 is four. **[2 marks]**

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6

Year t/Sales y ($1000s)		Two-point moving median	Two-point moving median with centring
1	10		
		Med (10,11) = 10.5	
2	11		11
		Med (11,12) = 11.5	
3	12		11.25
		Med (12,10) = 11	
4	10		11
		Med (10,12) = 11	
5	12		12.25
		Med (12,15) = 13.5	
6	15		

[3 marks]

Trend line: $y = 0.35t + 10.15$ **[1 mark]**

Sales for 2014: $t = 7$

$y = 2.4 + 10.15$

$= 12.6$

$\therefore$ \$12 600 in sales predicted for 2014 **[2 marks]**

Part C – Analysis questions

1 a

Day t	Temperature y (°F)	Three-point moving average
1	110	
2	120	(110 + 120 + 130)/3 = 120
3	130	(120 + 130 + 145)/3 = 131.67
4	145	(130 + 145 + 150)/3 = 141.67
5	150	(145 + 150 + 120)/3 = 138.33
6	120	(150 + 120 + 130)/3 = 133.33
7	130	(120 + 130 + 140)/3 = 130
8	140	(130 + 140 + 155)/3 = 141.67
9	155	(140 + 155 + 160)/3 = 151.67
10	160	(155 + 160 + 130)/3 = 148.33
11	130	(160 + 130 + 140)/3 = 143.33
12	140	(130 + 140 + 150)/3 = 140
13	150	(140 + 150 + 160)/3 = 150
14	160	(150 + 160 + 175)/3 = 161.67
15	175	(160 + 175 + 140)/3 = 158.33
16	140	

[3 marks]

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b Trend line: $y = 122.9 + 2.26t$

When $t = 17$, $y = 161$.

Therefore, the predicted temperature for day 17 is 161°F.

[2 marks]

c Smoothing the time series data using a four-point moving average gives the following.

Day t	Temperature y	Four-point moving average	Four-point moving average with centring
1	110		
2	120		
		$(110 + 120 + 130 + 145)/4 = 126.25$	
3	130		131.25
		$(120 + 130 + 145 + 150)/4 = 136.25$	
4	145		136.25
		$(130 + 145 + 150 + 120)/4 = 136.25$	
5	150		136.25
		$(145 + 150 + 120 + 130)/4 = 136.25$	
6	120		135.625
		$(150 + 120 + 130 + 140)/4 = 135$	
7	130		135.625
		$(120 + 130 + 140 + 155)/4 = 136.25$	
8	140		141.25
		$(130 + 140 + 155 + 160)/4 = 146.25$	
9	155		146.25
		$(140 + 155 + 160 + 130)/4 = 146.25$	
10	160		146.25
		$(155 + 160 + 130 + 140)/4 = 146.25$	
11	130		145.625
		$(160 + 130 + 140 + 150)/4 = 145$	
12	140		145
		$(130 + 140 + 150 + 160)/4 = 145$	
13	150		150.625
		$(140 + 150 + 160 + 175)/4 = 156.25$	
14	160		156.25
		$(150 + 160 + 175 + 140)/4 = 156.25$	
15	175		
16	140		

[4 marks]

d Trend is best described as seasonal. **[1 mark]**

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2

Year	Aces recorded set 1	Aces recorded set 2	Aces recorded set 3	Aces recorded set 4	Yearly average
2010	10/45	70/45	90/45	10/45	180/4 = 45
2011	10/55	80/55	90/55	40/55	220/4 = 55
2012	20/55	90/55	100/55	10/55	220/4 = 55
SI	0.26	1.55	1.82	0.38	

a

SI	0.26	1.55	1.82	0.38

[4 marks]

b

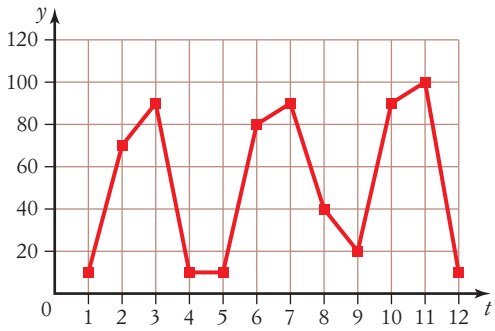
Year	Aces recorded set 1	Aces recorded set 2	Aces recorded set 3	Aces recorded set 4
2010	10/0.26 = 38.46	70/1.55 = 45.16	90/1.82 = 49.45	10/0.38 = 26.32
2011	10/0.26 = 38.46	80/1.55 = 51.61	90/1.82 = 49.45	40/0.38 = 105.26
2012	20/0.26 = 76.92	90/1.55 = 58.06	100/1.82 = 54.95	10/0.38 = 26.32

Deseasonalised data

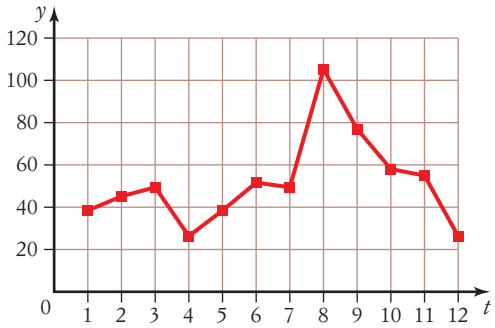
Year	Aces recorded set 1	Aces recorded set 2	Aces recorded set 3	Aces recorded set 4
2010	38.46	45.16	49.45	26.32
2011	38.46	51.61	49.45	105.26
2012	76.92	58.06	54.95	26.32

[4 marks]

c Original time series data



Deseasonalised time series data



[2 marks]

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